

A THREE-STEP MODEL OF BLACK SILICON FORMATION IN DEEP REACTIVE ION ETCHING PROCESS

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ABSTRACT

A three-step model used for modeling and simulation of black silicon formation in DRIE (Deep Reactive Ion Etching) process is presented. It divides the plasma etching system into plasma layer, sheath layer and sample surface layer. At the same time, it combines quantum mechanics, sheath dynamics and diffusion theory together based on plasma environment to predict the probability distribution of etching particles so as to simulate the final etching results. The simulation results show very good coincidence with experimental images, proving the applicability of this theory and it's promising to make black silicon formation in DRIE process to be controllable and repeatable.

INTRODUCTION

As microdevices scaling down to nanoscale, controllable formation of nanostructures by traditional MEMS process becomes a challenging technique. Plasma-based processes have attracted major attention owing to their outstanding simplicity, efficacy and cost-efficiency. Black silicon, named due to its impressive black surface, is a typical example of plasma-made nanostructures [1-2]. Taking advantage of its outstanding performance of low-reflectivity and super hydrophobicity, black silicon is experiencing quantum leaps in many application fields, such as solar cells, optical sensors, and microfluidics and so on [3-4].

The black silicon surface can be fabricated in several methods, which are always combinations of nanolithography and mask-assisted etching, like electron beam, focused ion beam (FIB) and normal Bosch DRIE (Deep Reactive Ion Etching) [5-6]. However, they usually have inherent limits, such as large structure size, low-efficient, low aspect ratio and uncontrollability. Unlike those mask-based fabrication methods, an optimized DRIE process which is maskless, low-cost, high-efficient, single-step and controllable, was employed to achieve black silicon [7-8].

In order to control the plasma-based black silicon process, the research on black silicon formation mechanism is becoming more and more significant. There have been many efforts from different research groups, such as investigating relationship between experimental process parameters and structure shape of black silicon, or analyzing chemical reactions related to the formation process [8-9]. However, there is little research on presenting the practical formation process of black silicon theoretically.

In this study, the etching process is described as probability event. A three-step model, combining quantum mechanics, sheath dynamics and diffusion theory, is developed to model the motion of etching particles, estimate the energy distribution on substrate and simulate the formation process of needle-like nanostructures.

OPTIMIZED DRIE PROCESS

According to Figure 1, DRIE process is usually carried out using $\text{SF}_6/\text{C}_4\text{F}_8$ in a mode of cyclic etching-passivation steps. Compared to normal DRIE process, there appear nanomasks after a few cycles in the optimized DRIE process, and they will work as the nanomasks during the following DRIE process, thus nanostructures are formed. It is widely believed that the generation of nanomasks is caused by the residual of polymeric layer deposited during passivation step [8]. Therefore, the generation of nanomasks should be the key point of investigating the formation mechanism of black silicon in plasma-based process.

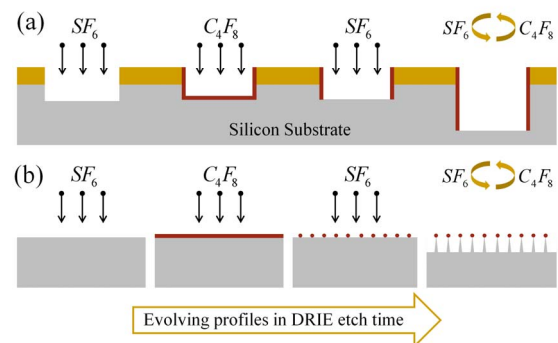


Figure 1: Schematic illustration of black silicon formation: (a) normal DRIE process, (b) optimized DRIE process.

Figure 2 shows the schematic of a typical dual source DRIE system which consist of an ICP (inductively coupled plasma) module and a CCP (capacitively coupled plasma) module [9]. High-density plasma glow is created by ICP source. Then the ions from plasma glow region are led by a RF (Radio Frequency) power towards the wafer surface with increasing ion energy. In fact, where RF power works is an electric field called sheath, which is between the plasma and the sample electrode [10]. Thus, the plasma etching system can be divided into three parts, i.e. plasma layer, sheath layer and sample surface layer. The etching particles enter from above, go through the plasma layer and sheath layer, then impinge the sample surface. There is distinct particle motion state in each part.

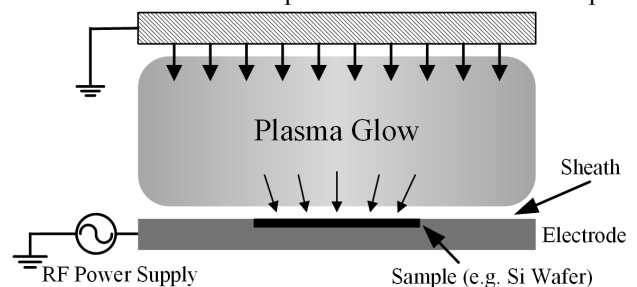


Figure 2: Schematic of a typical dual source DRIE system which consists of both coin module and plate module.

THREE-STEP MODEL FOR SIMULATION OF BLACK SILICON FORMATION

Figure 3 is a schematic illustration of the proposed three-step model. As mentioned above, the plasma etching system is divided into plasma layer, sheath layer and sample surface layer. Correspondingly, quantum mechanics, sheath dynamics and diffusion theory are adopted in each layer according to the movement characteristics of etching particles. A $6\mu\text{m} \times 6\mu\text{m}$ area in Figure 3 is the selected simulation area. These three layers are in sequence. The simulation result of last layer will become the initial value for the next layer. As passivation process and radical etching can be regarded as uniform chemical deposition or reaction, only the motion of ions during etching period should be taken into consideration.

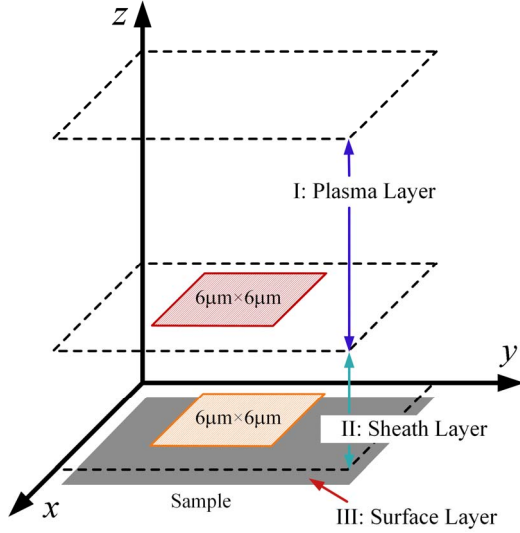


Figure 3: Schematic of three-step model: (I) plasma layer, (II) sheath layer and (III) sample surface layer.

Plasma Layer

Everything possesses wave property and it's particularly notable for microscopic particles. There exists uncertainty in the motion of object and the matter wave is also called probability wave. Analyzing shows the etching particles' motion during DRIE quite conforms to the typical motion elaborated in quantum mechanics [11-12]. So quantum mechanics is introduced here to estimate the probability that etching particles reach each point. In quantum mechanics, motion of particles can be described by Schrodinger equation:

$$i\hbar \frac{\partial \varphi_p}{\partial t} = - \left[\frac{\hbar^2}{2M} \nabla^2 + V(\vec{r}, t) \right] \varphi_p \quad (1)$$

Here, $V(\vec{r}, t)$ is field potential at position $\vec{r}$, M is mass of particles and $\hbar$ is Dirac constant. φ_p is function of matter wave, as shown in Equation (2):

$$\varphi_p = \varphi_p(\vec{r}, t) = A \exp\left(\frac{i(\vec{p} \cdot \vec{r} - Et)}{\hbar}\right) \quad (2)$$

where $\vec{p}$ and E are momentum and kinetic energy of particles, respectively.

According to theory of plasma sheath [10-11], $V(\vec{r}, t)$ is considered as zero in plasma layer. It's assumed

that there exist two principal momentum magnitudes for the incident particles of the order $10^{-20} \sim 10^{-21} \text{ kg} \cdot \text{m/s}$. After taking them into Equation (2) and (1), the probability distribution of etching particles on the simulation area (red area in Figure 3) can be achieved, as shown in Figure 4. It can be taken approximately as energy distribution because the probability and etching capability are of positive correlations. Meanwhile, this result also serves as initial value for the model in sheath layer. Thus, the difference in etch rate is caused by the uneven probability distribution of etching particles, moreover, the difference in etch rate is the principal contributor to self-masking effect.

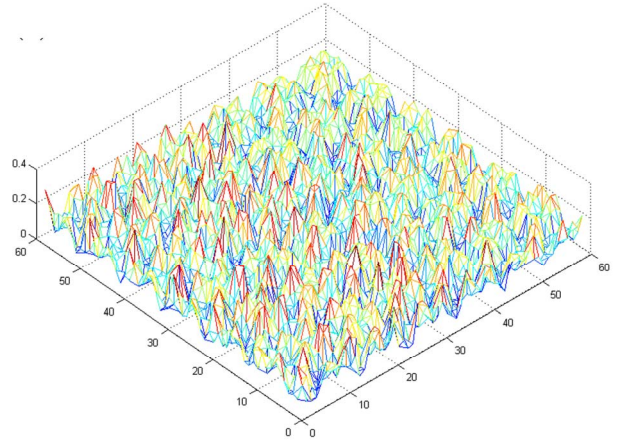


Figure 4: Simulated probability distribution on bottom surface of plasma layer

Sheath Layer

After leaving plasma layer, etching particles will go through the sheath layer before impinging on the sample. The electrode that the substrate is placed upon is often independently RF-biased to provide the electric fields within sheath layer and to increase the ions energy because the potential drop caused by RF bias voltage mainly happens in sheath layer. The quantum mechanics can also be used to describe the motion of etching particles in sheath layer. However, $V(\vec{r}, t)$ in Equation (1) is changed. It is nonlinear for this layer due to the sheath potential.

To calculate the sheath potential expression, a self-consistent model of sheath dynamics was used [10, 13]. It consists of equations describing the charge transport in sheath layer coupled to an equivalent circuit model to predict thickness of sheath layer, sheath potential and spatiotemporal variation of potential in sheath layer. All the physical quantities of sheath layer are considered to vary only in the vertical direction towards plate because it is much larger than that in horizontal direction.

Fig.5 shows spatiotemporal variation of potential in sheath layer. The curve oscillates over time in the direction as the blue arrows both in x-axis direction and y-axis direction indicate. The thickness of sheath layer equals to the distance from electrode to the point where potential value equals zero, i.e. the thickness of sheath layer equals to the x-value of point A. So the oscillation curve A shows time dependent sheath thickness calculated from the sheath model. Similarly, the potential on electrode equals to the y-value of point B. So the oscillation curve B shows time dependent potential on bottom electrode.

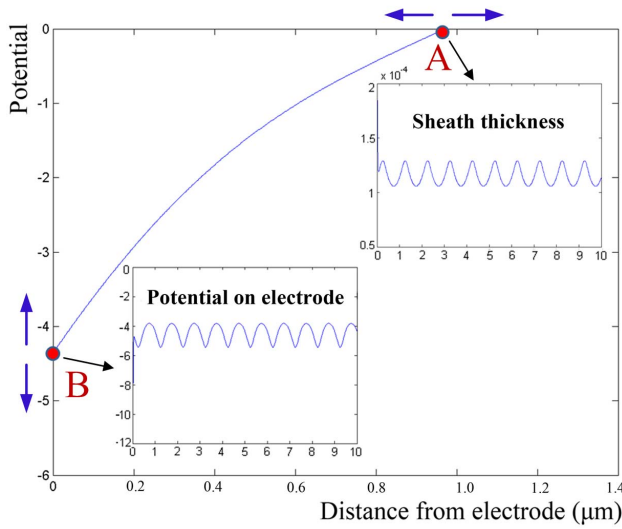


Figure 5: Simulated oscillation curves in sheath model: spatiotemporal variation of potential in sheath layer.

With the sheath potential derived from the above, Equation (1) can be solved by numerical iteration and the probability distribution on the simulation area in sheath layer (orange area in Figure 3) is shown in Figure 6. Compared to Figure 4, it's easy to notice that the sheath potential not only increases the maximum probability but also enhances concentration, making it easier for formation of high-aspect-ratio structures.

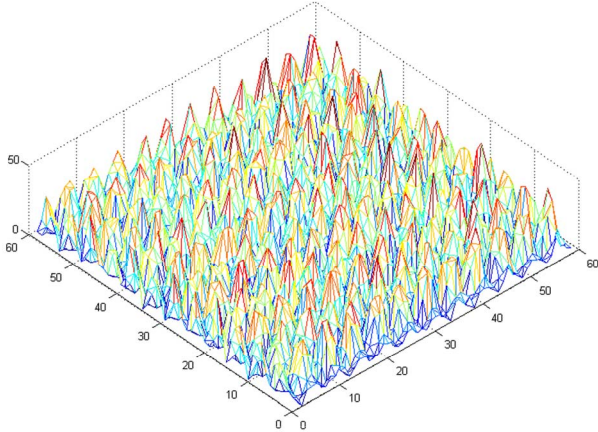


Figure 6: Simulated probability distribution on bottom surface of sheath layer.

Sample Surface Layer

When etching particles impinge on the sample surface, they are supposed to be affected by electric field generated by stored charge in the substrate. Two reasons are considered here: the particles are quite close to the charge on surface; the main factor that influences the final result in this layer is the plate-parallel motion of particles because sheath potential is considered to vary only in the vertical direction towards plate. A diffusion equation [14] shown below is used to model the process.

$$\frac{\partial h}{\partial t} = -D_h \nabla^2 h \quad (3)$$

Here, h is a variable of height which can be obtained after sheath layer simulation. D_h is diffusion coefficient.

According to above, the particles will diffuse from convex part to concave part on the surface. As a result, the concave part will receive more particles and be etched more.

RESULTS AND DISCUSSION

Finally, the simulated surface morphology can be obtained by rendering them in OpenDX. Furthermore, a matrix of simulated 3D images of nanostructures can be mapped out through varying momentum magnitude within reasonable range as is shown in Fig.7. Here, it is assumed that there exist two kinds of momentum magnitude in the experimental environment, the x-axis stands for one momentum magnitude while y-axis stands for the other. Under different combination of momentum magnitude, there will form nanostructures with different morphology. In this matrix, there are 45 simulated nanostructure morphologies which differ from one another. We believe for each of them, there should be a corresponding experimental sample with the same nanostructures, though we cannot realize all of them now subject to the experimental limitation.

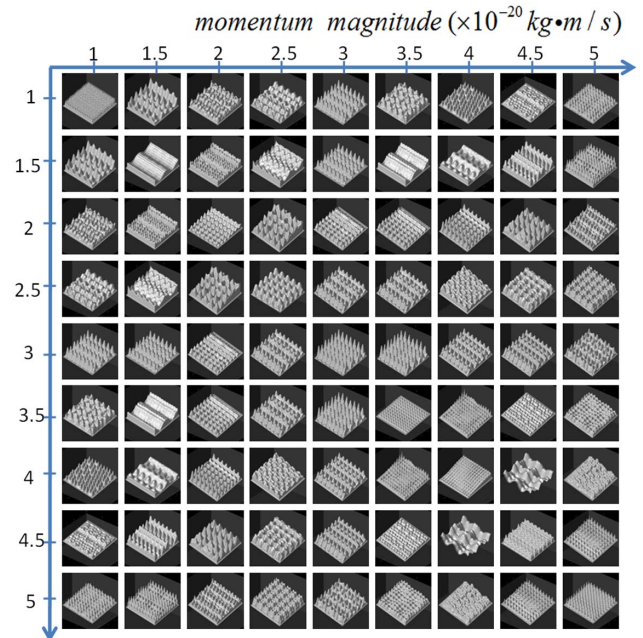


Figure 7: A matrix of 3D simulated nanostructures with reasonable range of momentum magnitude of etching particles (the matrix is symmetric).

In order to observe more clearly, four groups of comparison between simulation results and experimental SEM images of black silicon are listed in Fig.8. As we can see, they are consistent with each other both in profile and structure scale. Besides, they have the same distribution characteristic, such as some coterminous structures and the spacing between nanostructures. If we try to vary the momentum magnitude of etching particles in a larger range, it is promising to obtain more distinct nanostructure morphologies.

CONCLUSIONS

This paper focuses on the exploration of black silicon formation mechanism in DRIE process. It proposes a brand new simulation opinion in this field. Quantum mechanics,

sheath dynamics and diffusion theory are combined together to simulate the probability distribution of etching particles. This three-step model gives out very inspiring results to control and predict the fabrication process. It's worth further research to improve the model and establish a better theory system for plasma etching process.

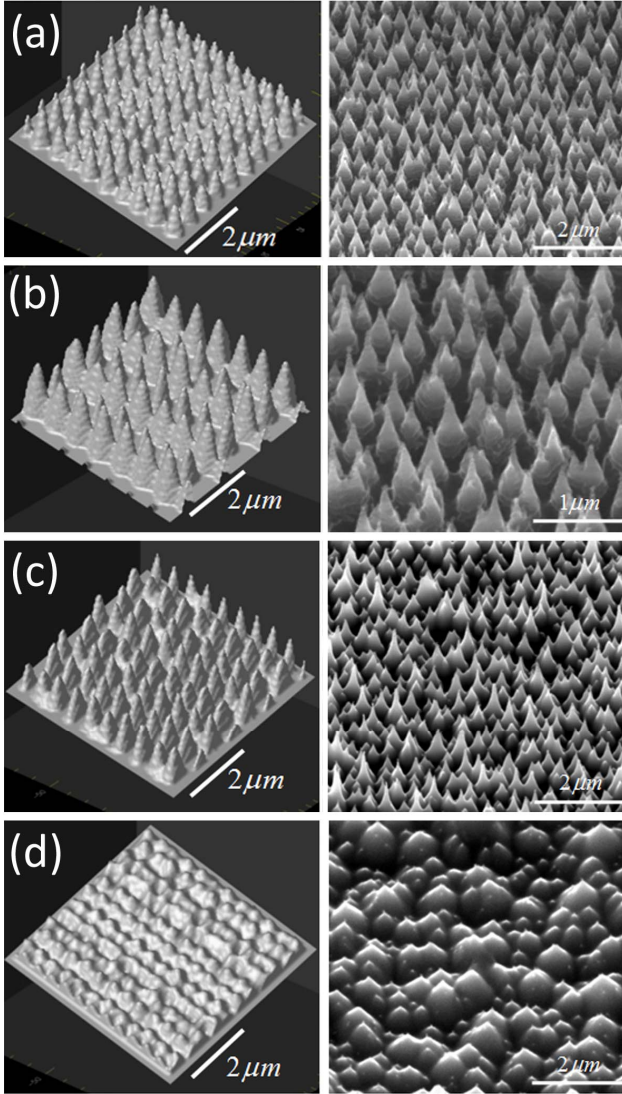


Figure 7: Comparisons between simulation result and SEM image of black silicon.

- (a) $|\overline{p_1}| = 4e - 20kg \cdot m/s, |\overline{p_2}| = 2e - 20kg \cdot m/s$;
(b) $|\overline{p_1}| = 3e - 20kg \cdot m/s, |\overline{p_2}| = 1e - 20kg \cdot m/s$;
(c) $|\overline{p_1}| = 4e - 20kg \cdot m/s, |\overline{p_2}| = 1e - 20kg \cdot m/s$;
(d) $|\overline{p_1}| = 4.5e - 20kg \cdot m/s, |\overline{p_2}| = 3.5e - 20kg \cdot m/s$.

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